COMP 4332 / RMBI 4310
Big Data Mining (Spring 2026)

Project 1 Emotion Classification

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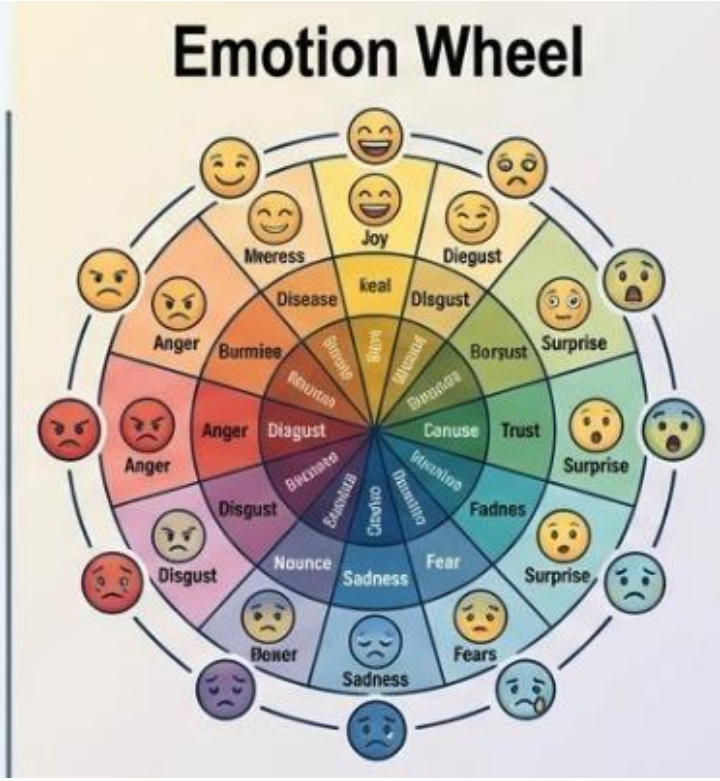
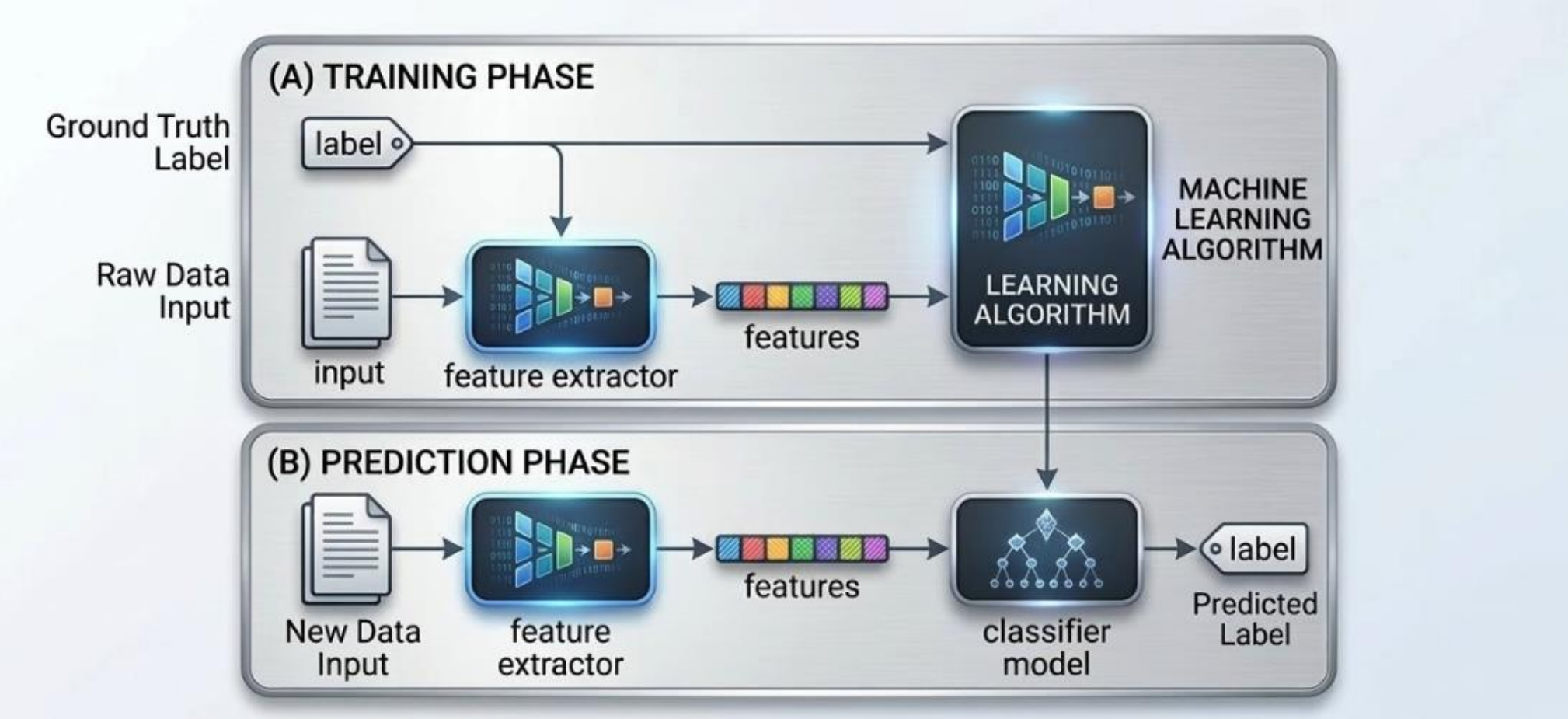
SENTIMENT VS. EMOTION



- Traditional **Sentiment Analysis** focuses on polarity (Positive, Negative, Neutral). However, human expression is far more complex.

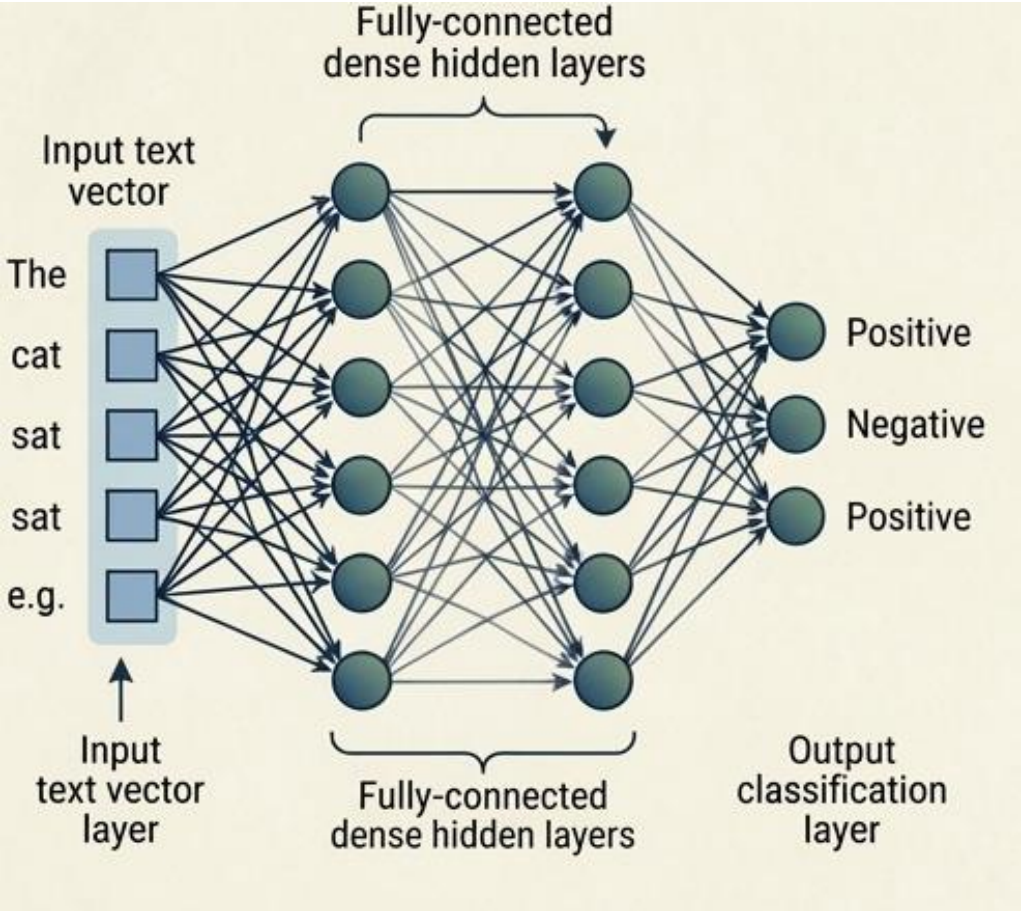
- **Emotion Recognition** aims to identify specific affective states, providing deeper insights into user intent, mental health, and social dynamics.

General Pipeline of Classification

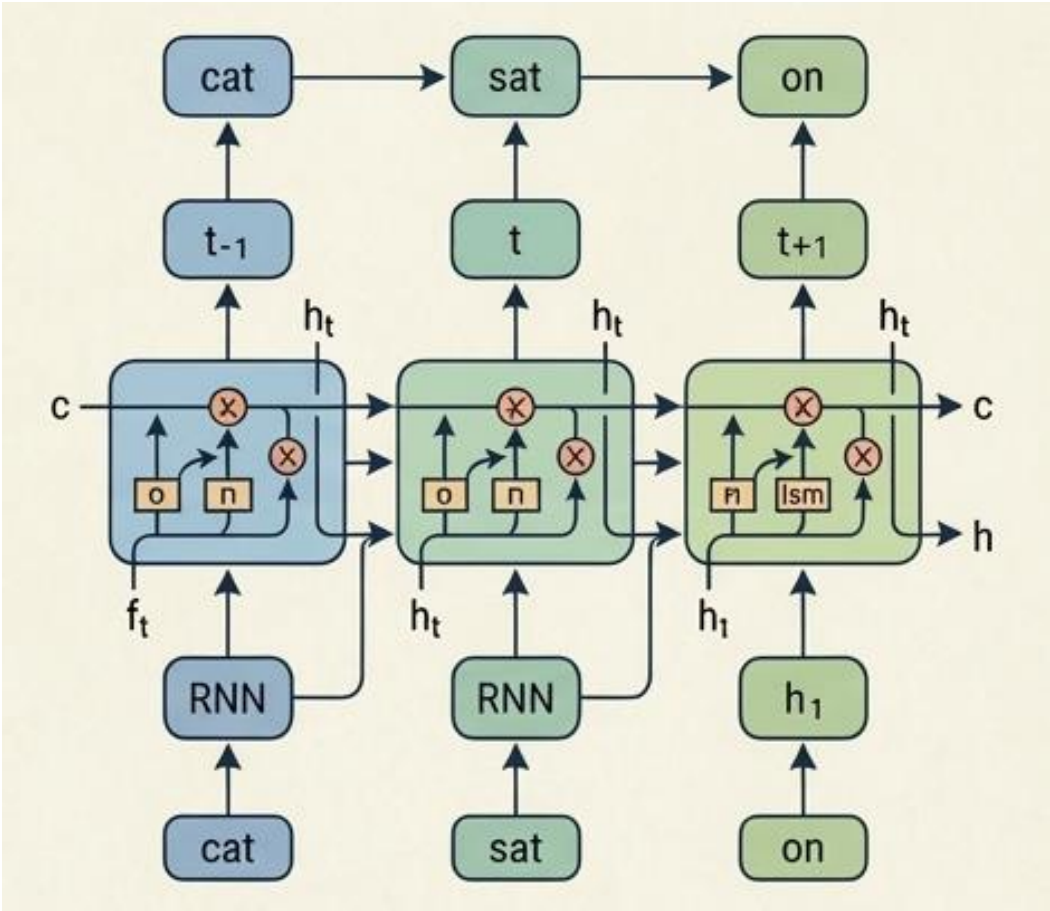


Text Classification with Deep Learning

MLP



RNN/LSTM



Implementations

Methods, frameworks and even programming languages are not restricted

Data preprocessing

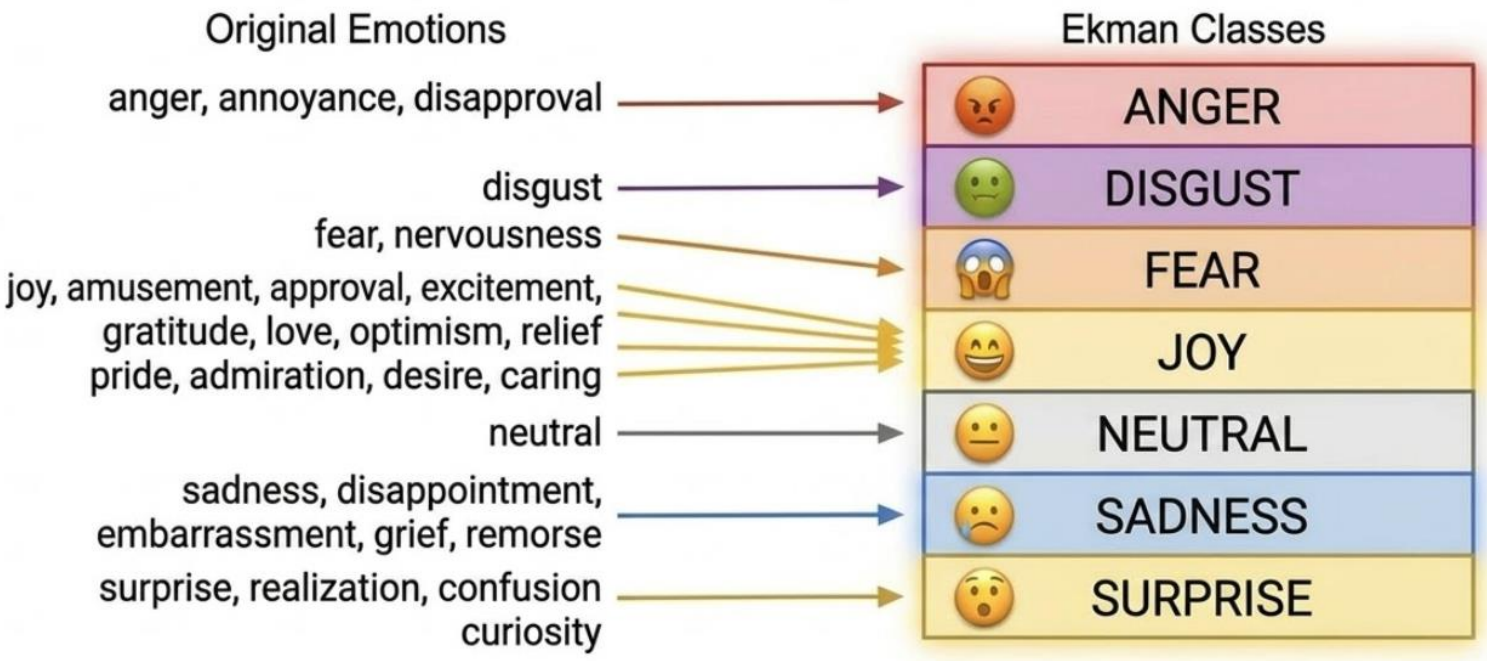
- Bag of words
- TF-IDF
- Word Embedding (train from scratch/load from pretrained word embeddings...)

Model

- Classical Machine learning: Naive Bayes, Logistic Regression, SVM...
- Neural networks: MLP, RNN/GRU/LSTM, TextCNN...
- Pretrained models: BERT, RoBERTa...

- Developed by Google Research, GoEmotions is a dataset of Reddit comments labeled with emotions.
- To simplify the task, the 27 original emotions are mapped to 7 Ekman categories using the official mapping.
- Only examples that map to exactly one Ekman class are kept. Multi-label examples are removed.

The GoEmotions Dataset



We provide: data

- ‘train.csv’ : 39.6k (id, text, label)
- ‘valid.csv’ : 4.9k (id, text, label)
- ‘test_no_label.csv’: 4.9k (id, text)

Labels are integers 0-6 in alphabetical order.

id	text	label
ee1lws5	I don't care about him at all. 😤😤😤	0
ed07q86	So sorry to hear that matey. What a good dog. ...	5

Integer	Emotion
0	anger
1	disgust
2	fear
3	joy
4	neutral
5	sadness
6	surprise

We provide: baselines with code base

Two baselines are provided: MLP and RNN

MLP

- Architecture: mean-pooled word embeddings with 3-layer MLP
- Input: tokenized text
- Embeddings: random, learned from scratch
- Output: 7-way classification

RNN

- Architecture: bidirectional vanilla RNN (tanh) with random embeddings learned from scratch
- Input: tokenized text
- Output: final hidden states concatenated -> linear layer -> 7-way classification

We provide: baselines with code base

For the code framework, the baselines share a common structure that you can reuse for your own models. It includes:

Data preprocessing

- Tokenization
- Vocabulary build
- Encoding
- Dataset class

Model Definition

Training Loop

Evaluation

Using the existing code framework isn't mandatory; you can also implement it yourself from scratch.

Submission

- Predict on **test data** and submit your predictions in a .csv file with `id` and `label` (please make sure you can successfully evaluate your validation predictions on the validation data with the help of evaluate.py)
- Report (2-3 pages)
- Code (Frameworks and even programming languages are not restricted.)
- **DDL: March 31st**
- Submission: Each **team leader** is required to submit the <GroupNo>.zip file with the structure below:

  <GroupNo>/
  - prediction.csv
  - report.pdf
  - Your implementation (scripts/notebooks/modules)

  To save space, do not include the original dataset files or model checkpoints in your submission.

- We will check your report with your code and the Macro-F1.

Grading Rule

Grade	Model (80%)	Report (20%)	Baseline (Macro-f1 on test set)
60%		submission	0.44
80%	an easy baseline that most students can outperform	detailed explanation	0.46
90%	a competitive baseline that about half students can surpass	detailed explanation and analysis	0.50
100%	a very competitive baseline	excellent visualization and analysis	0.52

Thank You